

PARTIAL DIFFERENTIAL EQUATIONS

UNIT III – COMPLETE FORMULA & REVISION SHEET

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COURSE OBJECTIVES

- Understand **formation** of PDEs from relations
- Eliminate arbitrary **constants** & **functions**
- Solve first-order PDEs using **Lagrange's method**
- Solve **homogeneous linear PDEs** with constant coefficients
- Find **Complementary Function (CF)** and **Particular Integral (PI)**
- Apply PDEs to basic **physical problems**

BASIC DEFINITIONS

PDE

An equation involving **partial derivatives** of a dependent variable w.r.t. two or more independent variables.

ORDER

The order of the **highest** partial derivative present in the PDE.

DEGREE

The power of the **highest order** derivative after clearing fractions/radicals.

DEPENDENT VARIABLE

The variable being differentiated (usually z or u).

INDEPENDENT VARIABLES

Variables w.r.t. which we differentiate (usually x, y).

LINEAR PDE

Dependent variable and its derivatives appear **linearly** (degree 1, no products).

NON-LINEAR PDE

Contains **products, powers, or non-linear terms** of the dependent variable / its derivatives.

HOMOGENEOUS PDE

All derivatives are of the **same order** and $\text{RHS} = 0$.

GENERAL SOLUTION

Contains **arbitrary functions** (most general); satisfies the PDE.

COMPLETE / PARTICULAR SOLUTION

Complete: has as many arbitrary constants as the order.
Particular: specific values assigned.

Integral Surface: The surface $z = f(x, y)$ that satisfies the PDE — the geometric interpretation of its solution.

STANDARD NOTATION

$$p = \partial z / \partial x$$

$$q = \partial z / \partial y$$

$$D = \partial / \partial x, D' = \partial / \partial y$$

$$r = \partial^2 z / \partial x^2$$

$$s = \partial^2 z / \partial x \partial y$$

$$t = \partial^2 z / \partial y^2$$

FORMATION OF PDES

METHOD 1 – ELIMINATION OF ARBITRARY CONSTANTS

Used when the relation has **arbitrary constants** (a, b, c, ...)

- 1 Write the given relation: $f(x, y, z, a, b) = 0$
- 2 Differentiate partially w.r.t. x (keep y constant) → get equation with p
- 3 Differentiate partially w.r.t. y (keep x constant) → get equation with q
- 4 Eliminate the constants a, b from the three equations
- 5 Result is the PDE (should have no arbitrary constants)

Key Rule: Number of steps of differentiation = Number of arbitrary constants to eliminate

METHOD 2 – ELIMINATION OF ARBITRARY FUNCTIONS

Used when the relation has **arbitrary functions** $\phi, f, g, \dots$

- 1 Write: $z = f(u)$ or $f(u, v) = 0$ where u, v are expressions in x, y, z
- 2 Differentiate w.r.t. $x \rightarrow$ introduces $f'(u)$ with p
- 3 Differentiate w.r.t. $y \rightarrow$ introduces $f'(u)$ with q
- 4 Divide or manipulate to **eliminate** $f'(u)$
- 5 Result is the PDE containing p and q

Key Rule: One arbitrary function → 1st order PDE; Two functions → 2nd order PDE

FORMATION METHODS – COMPARISON

FEATURE	ELIMINATION OF CONSTANTS	ELIMINATION OF FUNCTIONS
Given	Relation with $a, b, c, \dots$	Relation with $f(\cdot), \phi(\cdot), \dots$
Step Count	= No. of constants	Usually 2 differentiations
Tools Used	$\partial/\partial x, \partial/\partial y$, then algebra	Chain rule, then eliminate f'
Order of PDE	Usually 1st order	1st or 2nd order
Example Form	$z = ax + by + ab$	$z = f(x^2 + y^2)$

FIRST ORDER PDES

GENERAL FIRST ORDER PDE

$$F(x, y, z, p, q) = 0$$

LINEAR FIRST-ORDER PDE

$Pp + Qq = R$, where P, Q, R are functions of x, y, z . (p, q appear linearly)

QUASI-LINEAR

Linear in p and q , but P, Q, R may involve z .

NON-LINEAR

p and q appear with powers or products.

STANDARD LAGRANGE FORM

$$Pp + Qq = R$$

$$P = P(x, y, z)$$

$$Q = Q(x, y, z)$$

$$R = R(x, y, z)$$

General solution: $f(u, v) = 0$

GEOMETRIC MEANING

The solution surface $z = z(x, y)$ must be tangent to the vector field (P, Q, R) at every point.

LAGRANGE'S LINEAR EQUATION – COMPLETE METHOD

$$Pp + Qq = R \quad \rightarrow \quad \text{Auxiliary Equations:} \quad dx/P = dy/Q = dz/R$$

SOLUTION STEPS

- 1 Identify P, Q, R from the given PDE $Pp + Qq = R$
- 2 Write the **Auxiliary (Lagrange's) equations**: $dx/P = dy/Q = dz/R$
- 3 Solve any **two pairs** of the ratio equations to get two independent solutions: $u(x, y, z) = c_1$ and $v(x, y, z) = c_2$
- 4 Write the **General Solution**: $f(u, v) = 0$ or equivalently $u = \psi(v)$

General Solution Form:

$$f(u, v) = 0 \quad \text{or} \quad u = \psi(v)$$

where $u = c_1$ and $v = c_2$ are two independent first integrals.

TECHNIQUES FOR SOLVING AUXILIARY EQUATIONS

METHOD OF GROUPING

Take any two ratios that can be solved directly. Often $dx/P = dy/Q$ alone gives one integral.

METHOD OF MULTIPLIERS

Choose multipliers l, m, n (functions of x, y, z) such that:

$$lP + mQ + nR = 0$$

Then $l \cdot dx + m \cdot dy + n \cdot dz = 0$ is integrable $\rightarrow$ gives one integral.

Common Multiplier Sets:

$$l = x, m = y, n = z$$

$$l = 1/P, m = 1/Q, n = 1/R$$

$$l = y, m = x, n = 0$$

⚠ Remember: Two independent solutions u and v are needed. Never take two proportional integrals.

SHORTCUT STRATEGY FOR COMMON PDE TYPES

PDE FORM	P	Q	R	HINT FOR AUXILIARY EQS
$yp - xq = 0$	y	$-x$	0	$dx/y = dy/-x \rightarrow xdx + ydy = 0 \rightarrow x^2 + y^2 = c_1$; $dz/0 \rightarrow z = c_2$
$xp + yq = z$	x	y	z	$dx/x = dy/y \rightarrow y/x = c_1$; $dy/y = dz/z \rightarrow z/y = c_2$
$p + q = 1$	1	1	1	$dx = dy \rightarrow y - x = c_1$; $dx = dz \rightarrow z - x = c_2$
$(y-z)p + (z-x)q = x-y$	$y-z$	$z-x$	$x-y$	Use multipliers $1, 1, 1$: $x+y+z = c_1$; use x, y, z : $x^2 + y^2 + z^2 = c_2$

LAGRANGE VS HOMOGENEOUS PDE – KEY DIFFERENCES

FEATURE	LAGRANGE'S METHOD	HOMOGENEOUS LINEAR PDE
Order	1st order	2nd or higher order
Form	$Pp + Qq = R$	$f(D, D')z = F(x, y)$
Coefficients	Functions of x, y, z	Constant coefficients
Auxiliary Eq	$dx/P = dy/Q = dz/R$	$f(m, 1) = 0$
Solution Form	$f(u, v) = 0$	$z = CF + PI$
Key Technique	Two first integrals	Operator method

HOMOGENEOUS LINEAR PDE (CONSTANT COEFFICIENTS)

General Form: $f(D, D')z = F(x, y)$ where $D = \partial/\partial x$, $D' = \partial/\partial y$

STANDARD FORM (2ND ORDER)

$$(aD^2 + bDD' + cD'^2)z = F(x, y)$$

AUXILIARY EQUATION

- 1 Replace $D \rightarrow m$ and $D' \rightarrow 1$
- 2 Solve: $f(m, 1) = 0 \rightarrow$ get roots $m_1, m_2, \dots$
- 3 Write CF using roots

Auxiliary Equation (2nd order):

$$am^2 + bm + c = 0$$

COMPLETE SOLUTION

$$z = CF + PI$$

COMPLEMENTARY FUNCTION (CF) - ROOT TYPES

ROOT TYPE	CF FORM
Distinct real $m_1 \neq m_2$	$\psi_1(y + m_1x) + \psi_2(y + m_2x)$
Repeated real $m_1 = m_2 = m$	$\psi_1(y + mx) + x \cdot \psi_2(y + mx)$
Three distinct m_1, m_2, m_3	$\psi_1(y+m_1x) + \psi_2(y+m_2x) + \psi_3(y+m_3x)$
Triple repeat $m_1=m_2=m_3=m$	$\psi_1(y+mx) + x \cdot \psi_2(y+mx) + x^2 \cdot \psi_3(y+mx)$

PARTICULAR INTEGRAL (PI)

Concept: PI is any one particular solution of the non-homogeneous equation $f(D, D')z = F(x, y)$. It is found using the operator method.

PI OPERATOR METHOD

$$PI = [1 / f(D, D')] \cdot F(x, y)$$

CASE 1 - RHS = $E^{(AX+BY)}$

$$PI = e^{(ax+by)} / f(a, b)$$

△ Valid when $f(a, b) \neq 0$

CASE 2 - RHS = $\sin(AX+BY)$ OR $\cos(AX+BY)$

$$\text{Replace } D^2 \rightarrow -a^2, D'^2 \rightarrow -b^2, DD' \rightarrow -ab$$

After substitution, compute the resulting constant; if denominator is 0, use failure rule.

CASE 3 - RHS = $x^m y^n$ (POLYNOMIAL)

$$PI = [f(D, D')]^{-1} \cdot x^m y^n$$

Expand operator as binomial series, then operate term by term on $x^m y^n$.

CASE 4 - RHS = $E^{(AX+BY)} \cdot V(X, Y)$

$$PI = e^{(ax+by)} \cdot [1/f(D+a, D'+b)] \cdot V$$

FAILURE CASE & MODIFICATION RULE

When $f(a, b) = 0$:

The denominator becomes 0. Then:

$$PI = x \cdot [1 / (\partial f / \partial D)] \cdot e^{(ax+by)}$$

(Differentiate denominator w.r.t. D and multiply by x)

If still 0, multiply by x again (repeat).

KEY OPERATOR RULES

OPERATION	RULE
$D \cdot x^n$	$n \cdot x^{n-1}$
$D' \cdot y^n$	$n \cdot y^{n-1}$
$1/D \cdot f(x)$	$\int f(x)dx$ (integrate)
$1/D' \cdot f(y)$	$\int f(y)dy$ (integrate)
$D^n \cdot e^x$	e^x (any n)

PROPERTY	COMPLEMENTARY FUNCTION (CF)	PARTICULAR INTEGRAL (PI)
Definition	Solution of $f(D,D')z = 0$	Any one solution of $f(D,D')z = F(x,y)$
Contains	Arbitrary functions $\phi_1, \phi_2, \dots$	No arbitrary functions
How found	Auxiliary equation $\rightarrow$ roots	Operator method
Role	Represents the homogeneous part	Accounts for RHS forcing



FLASHCARDS – RAPID FIRE REVISION

Cover the answer. Test yourself. Repeat until perfect.

Q: What is a PDE? Equation with partial derivatives of a dependent variable w.r.t. 2+ independent variables	Q: Order of a PDE? Order of the HIGHEST partial derivative present	Q: Degree of a PDE? Power of the highest order derivative (after clearing fractions)
Q: p and q represent? $p = \partial z / \partial x$ $q = \partial z / \partial y$	Q: Lagrange equation form? $Pp + Qq = R$	Q: Auxiliary equations for Lagrange? $dx/P = dy/Q = dz/R$
Q: General solution of Lagrange's eq? $f(u, v) = 0$ or $u = \psi(v)$	Q: What are u and v in Lagrange? Two independent first integrals ($u=c_1$, $v=c_2$)	Q: Operator D and D'? $D = \partial/\partial x$ $D' = \partial/\partial y$
Q: Homogeneous PDE condition? All partial derivative terms are of the SAME order; RHS = 0 (for CF)	Q: Auxiliary equation for homogeneous PDE? Replace $D \rightarrow m$, $D' \rightarrow 1 \rightarrow$ solve $f(m, 1) = 0$	Q: CF for distinct real roots m_1, m_2? $\psi_1(y+m_1x) + \psi_2(y+m_2x)$
Q: CF for repeated root m (x2)? $\psi_1(y+mx) + x \cdot \psi_2(y+mx)$	Q: Complete solution of non-homogeneous? $z = CF + PI$	Q: PI when RHS = $e^{(ax+by)}$? $PI = e^{(ax+by)} / f(a, b)$ [if $f(a, b) \neq 0$]
Q: PI failure case? When $f(a, b) = 0 \rightarrow$ multiply by x, differentiate denominator w.r.t. D	Q: For sin/cos PI: what to substitute? $D^2 \rightarrow -a^2$, $D'^2 \rightarrow -b^2$, $DD' \rightarrow -ab$	Q: PI for polynomial RHS? Expand $[f(D, D')^{-1}]$ as binomial series, operate on $x^m y^n$ term by term
Q: Method of multipliers: condition? Choose l, m, n such that $lP + mQ + nR = 0$	Q: Formation by constants: key step? Differentiate twice (once w.r.t. x, once w.r.t. y) then eliminate constants	Q: Formation by functions: key step? Use chain rule to differentiate, then eliminate the arbitrary function $f'(u)$
Q: Integral surface means? The surface $z=f(x, y)$ satisfying the PDE – geometric solution interpretation	Q: 1/D operating on f(x)? $\int f(x) dx$ (integrate w.r.t. x)	Q: Linear PDE definition? Dependent variable and all its derivatives appear to the FIRST power only
Q: How many first integrals needed in Lagrange? EXACTLY TWO independent integrals: $u=c_1$ and $v=c_2$	Q: General solution has...? Arbitrary FUNCTIONS (not constants); satisfies the PDE completely	Q: r, s, t notation? $r = \partial^2 z / \partial x^2$, $s = \partial^2 z / \partial x \partial y$, $t = \partial^2 z / \partial y^2$

⚠ EXAM TIPS – COMMON MISTAKES TO AVOID

- ⚠ Confusing ODE and PDE: In ODE, only one independent variable; in PDE, **two or more**. Never treat PDE like an ODE.
- ⚠ Forgetting to eliminate **both** constants when forming a PDE – one differentiation per constant.
- ⚠ Not writing the auxiliary equations correctly – write **$dx/P = dy/Q = dz/R$** fully before solving.
- ⚠ Selecting two **dependent** integrals in Lagrange's method (e.g., both from x and y only) – always ensure independence.
- ⚠ Forgetting the **arbitrary function** in the final answer of Lagrange's method – result must be **$f(u, v) = 0$** .
- ⚠ Using **wrong substitution** in PI for sin/cos – must replace $D^2 \rightarrow -a^2$, not $D \rightarrow ia$.
- ⚠ Missing the **failure rule** when $f(a, b) = 0$ for exponential PI – don't leave answer undefined.
- ⚠ Wrong **sign** in differentiation – be careful: $\partial(-x^2)/\partial x = -2x$, not $+2x$.
- ⚠ Forgetting that **CF for repeated roots** requires multiplication by x for each repetition.
- ⚠ Writing PI with arbitrary functions – **PI has NO arbitrary functions**; only CF does.

TYPES OF PDE – MASTER TABLE

TYPE	CHARACTERISTIC	EXAMPLE	SOLUTION METHOD
Linear, 1st Order	p, q appear linearly; no products	$Pp + Qq = R$	Lagrange's method
Non-linear, 1st Order	p or q has power >1 or product pq	$p^2 + q^2 = 1$	Charpit's / special methods
Homogeneous, 2nd Order	All terms same order; const. coeff.; RHS=0	$(D^2 - DD')z = 0$	CF using auxiliary eq.
Non-Homogeneous, 2nd Order	Const. coeff.; RHS = $F(x, y) \neq 0$	$(D^2 - D'^2)z = e^{(x+y)}$	$z = CF + PI$
Wave Equation	Hyperbolic; $\partial^2 u / \partial t^2 = c^2 \partial^2 u / \partial x^2$	Vibrating string	Separation of variables
Heat Equation	Parabolic; $\partial u / \partial t = k \partial^2 u / \partial x^2$	Heat conduction	Separation of variables
Laplace Equation	Elliptic; $\partial^2 u / \partial x^2 + \partial^2 u / \partial y^2 = 0$	Steady-state temp.	Separation of variables

TYPES OF SOLUTIONS – SUMMARY

SOLUTION TYPE	DEFINITION	HAS ARBITRARY...	EXAMPLE
General Solution	Satisfies the PDE; most comprehensive	Functions of integration	$f(x-y, z-x) = 0$
Complete Solution	Contains n arbitrary constants (for order n)	Constants a, b, c	$z = ax + by + f(a, b)$
Particular Solution	Obtained by assigning specific values to constants/functions	Nothing (fully specified)	$z = 2x + 3y + 6$
Singular Solution	Cannot be derived from general solution by assigning constants	Nothing	Envelope of family of surfaces

QUICK FORMULA REFERENCE

ALL KEY FORMULAS • NO EXPLANATIONS • EXAM READY

NOTATION

SYMBOL	MEANING
p	$\partial z / \partial x$
q	$\partial z / \partial y$
r	$\partial^2 z / \partial x^2$
s	$\partial^2 z / \partial x \partial y$
t	$\partial^2 z / \partial y^2$
D	$\partial / \partial x$
D'	$\partial / \partial y$

FORMATION

GOAL	ACTION
Eliminate constants	$\partial / \partial x, \partial / \partial y$ then algebra
Eliminate functions	Chain rule $\rightarrow$ eliminate $f'(u)$
n constants	n differentiations needed

LAGRANGE'S METHOD

$$Pp + Qq = R$$

$$dx/P = dy/Q = dz/R$$

$$\text{General solution: } f(u, v) = 0$$

$$u = c_1 \text{ (first integral)} \quad | \quad v = c_2 \text{ (second integral)}$$

METHOD OF MULTIPLIERS

Choose l, m, n : $lP + mQ + nR = 0$
 Then: $l \cdot dx + m \cdot dy + n \cdot dz = 0 \rightarrow$ integrate

MASTER FORMULA STRIP

$$Pp + Qq = R$$

Lagrange Form

$$dx/P = dy/Q = dz/R$$

Auxiliary Eqs

$$f(u, v) = 0$$

Gen. Solution

$$f(m, 1) = 0$$

Aux. Eq. (Hom.)

$$z = CF + PI$$

Complete Solution

$$PI = e^{ax+by} / f(a, b)$$

Exponential PI

Remember the 3 Golden Rules: (1) Always write auxiliary equations before solving. (2) General solution needs TWO independent integrals. (3) Complete solution = CF + PI (never miss either part).

HOMOGENEOUS PDE - CF

ROOTS OF $F(m, 1) = 0$	CF
m_1, m_2 distinct	$\psi_1(y+m_1x) + \psi_2(y+m_2x)$
$m_1 = m_2 = m$ ($\times 2$)	$\psi_1(y+mx) + x \cdot \psi_2(y+mx)$
$m_1 = m_2 = m_3 = m$ ($\times 3$)	$\psi_1(y+mx) + x \cdot \psi_2(y+mx) + x^2 \cdot \psi_3(y+mx)$
Three distinct	$\psi_1(y+m_1x) + \psi_2(y+m_2x) + \psi_3(y+m_3x)$

PARTICULAR INTEGRAL (PI)

RHS = $F(x, y)$	PI FORMULA
e^{ax+by}	$e^{ax+by} / f(a, b) \quad [f(a, b) \neq 0]$
e^{ax+by} [fail]	$x \cdot e^{ax+by} / f'(a, b)$
$\sin(ax+by)$	Sub $D^2 \rightarrow -a^2, D'^2 \rightarrow -b^2, DD' \rightarrow -ab$
$\cos(ax+by)$	Same substitution as sin
$x^m y^n$	Binomial expansion of operator $^{-1}$
$e^{ax+by} \cdot V$	$e^{ax+by} \cdot [1/f(D+a, D'+b)] \cdot V$

OPERATOR RULES

EXPRESSION	RESULT
$1/D \cdot x^n$	$x^{n+1} / (n+1)$
$1/D' \cdot y^n$	$y^{n+1} / (n+1)$
$D^k \cdot e^{ax+by}$	$a^k \cdot e^{ax+by}$
$D'^k \cdot e^{ax+by}$	$b^k \cdot e^{ax+by}$

